
CONTEXTUAL LEARNING FOR INTERRUPTION TOLERANCE AND MULTITASKING

Thomas E. Portegys, portegys@gmail.com, ORCID 0000-0003-0087-6363

Dialectek, DeKalb, Illinois USA

ABSTRACT

This project is an effort to combine the Morphognosis and Mona neural network models into a comprehensive model for learning and behavior called Mandala. Mona features a contextual causation learning with goal-directed motivation. Morphognosis features contextual multilayer perceptron (MLP) learning. Mandala achieves this by externally accumulating tiers of temporal states that are fed into an MLP at each time step. Natural environments abound in event streams that require interruption tolerance and multitasking. Mandala affords multitasking as it is robust in the presence of intervening events from overlaid causation chains, a capability that conventional recurrent artificial neural networks (RNNs) struggle with. The results bear this out. In addition, externally accumulating temporal information discretely labels hierarchical cause-and-effect relationships that can be used for augmented processing. In the case of Mona, channeling motivation through the network for the purpose of goal-seeking requires this feature.

Keywords: Mandala, causation learning, multitasking, artificial neural network, machine learning.

INTRODUCTION

A mated pair of eagles, Jackie and Shadow, are popular on the internet these days as viewed with a hidden camera and microphone (Forgione, 2020). They spend their days and nights building and repairing their nest, brooding eggs, hunting, and fending off threats. In one segment Shadow is seen screaming for Jackie to help with intruding ravens that have in the past destroyed their eggs. He knows he must remain sitting atop them. She has been gone overlong and he must be hungry. This in general is a typical condition for animals and the neural networks that orchestrate them. Multitasking in the midst of interruptions and interferences between goals.

Artificial neural networks (ANNs) currently deal with what might be termed controlled surprise. Self-driving cars are in the focused business of driving. They can handle unexpected road obstructions which they have previously seen in training. Large Language Models (LLMs) (Yin, et al., 2024) can sensibly respond to just about any query, but while composing said response they do not deal with a crying baby or a flat tire. Human animals do.

This project is an effort to find a way how interruptions can be accommodated and multitasking can be accomplished more effectively by an ANN. The resulting architecture is named Mandala, after a geometric pattern that is typically composed of concentric rings or tiers of symbols having some comprehensive meaning (Merriam–Webster Online Dictionary, 2008) Here Mandala is also a tiered construction, where higher tiers represent longer-term context that aids in prediction.

Mandala is also a step toward synthesizing the Morphognosis (Portegys, 2022) and Mona (Portegys, 2010 and 2024b) neural network models into a comprehensive model for learning and behavior. Mona features a contextual causation learning with goal-directed motivation. Morphognosis features contextual multilayer perceptron (MLP) learning. Mandala achieves this by externally accumulating tiers of temporal states that are fed into an MLP at each time step. In addition, externally accumulating non-hidden temporal information discretely labels hierarchical cause-and-effect relationships that can be used for augmented processing. In the case of Mona, channeling motivation through the network for the purpose of goal-seeking requires this feature.

Recurrent Neural Networks (RNNs) with external temporal memory are architectures designed to overcome the limitations of standard RNNs (like LSTMs or GRUs) in memorizing long-term dependencies. While standard RNNs rely on internal hidden states that often suffer from vanishing gradients over long sequences, these models integrate a separate, addressable memory bank to store and retrieve past information over extended time horizons.

Neural Turing Machines (NTMs) (Graves et al., 2014). This is a model that couples a neural network controller to an external memory bank, capable of learning to manipulate and manage the memory to solve tasks like sorting or copying.

A Dynamic Memory Network (DMN) (Kumar et al., 2016) is a neural network architecture that uses an external episodic memory to answer questions about input sequences (stories, text).

External Addressable Long-Term and Working Memory (EALWM) (Quan et al., 2020) splits external memory into distinct long-term and working memory parts, improving the ability to learn long-term dependencies without gradient issues.

The above models retain an RNN at their core, using external storage as an adjunct memory. In Mandala external memory serves as the sole repository of temporal state, combining environment and state information as input to a non-recurrent MLP.

DESCRIPTION

Cause-effect hierarchy used to model maze learning, pufferfish, honey bees, and nesting birds. Is there a rationale for this? Here causation can be seen as prediction. Allows modularity and

prediction by context. This project assumes hierarchical, but could also work with a flat linear chain as well. Trying to fit into Mona scheme.

Predictive coding.

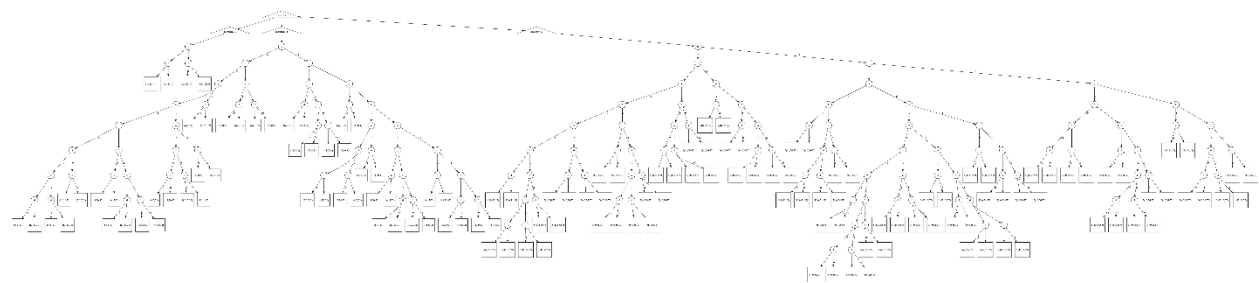
Aristotle

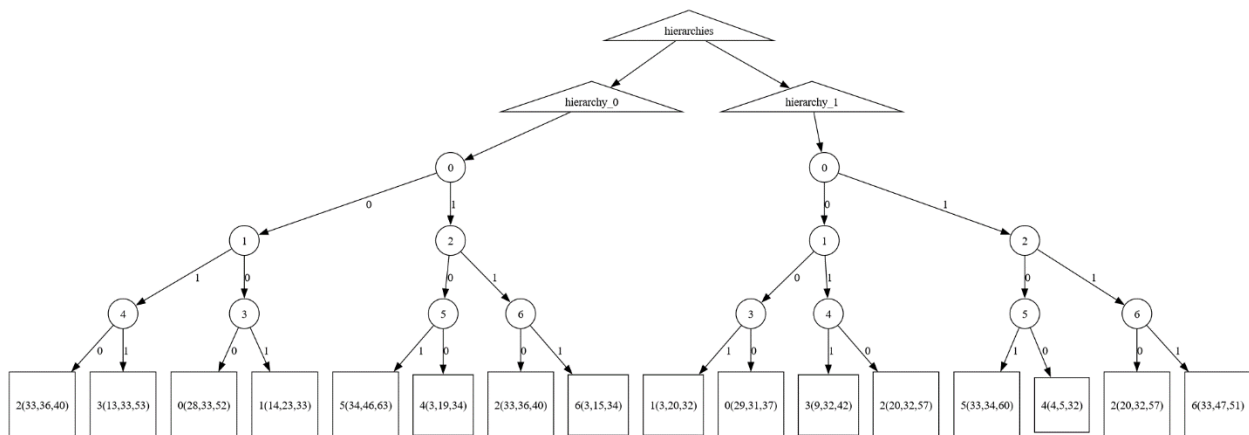
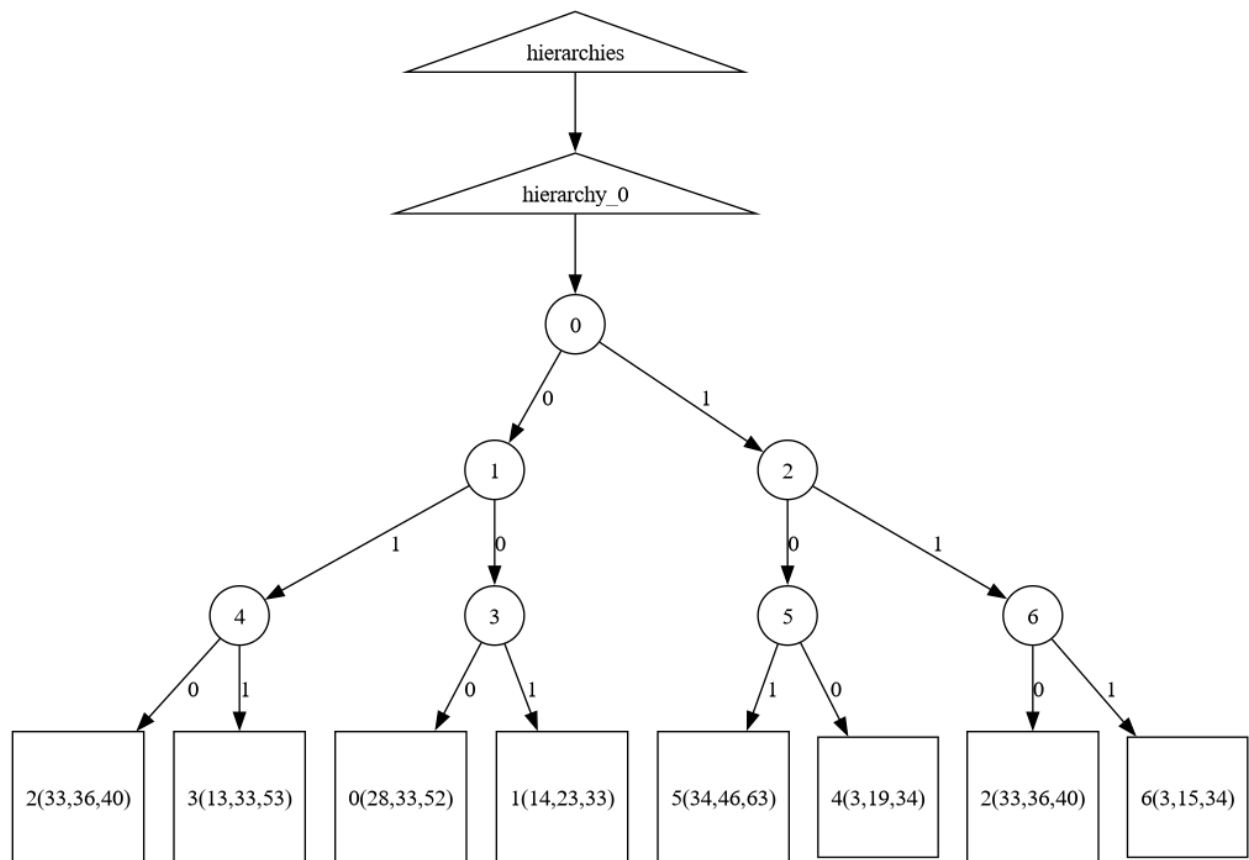
[John Stuart Mill](#) describes the Law of Universal Causation in following way:

Every phenomenon has a cause, which it invariably follows; and from this are derived other invariable sequences among the successive stages of the same effect, as well as between the effects resulting from causes which invariably succeed one another. ^[8]

System of Logic 1872, Book III p. 112.

[De Pierris, Graciela](#); [Friedman, Michael \(2008-06-04\)](#), [Kant and Hume on Causality in: Stanford Encyclopedia of Philosophy Archive](#), Metaphysics Research Lab, Stanford University





RESULTS

Tried Claude, Gemini, ChatGPT on test_0 sequences.

Given these sequences:

0,1,2,3

4,5,2,6

Predict the next number in these sequences:

8,8,7,0

7,11,4

7,4,5,11,11,8,10,10,10,11,2

0,1,9,11,10,2

7,9,11,9,7,8,4,5,7,10,10,8,7,2

7,0,1,11,10,7,7,10,2

4,5,10,8,8,2

11,8,0,1,8,10,2

11,7,7,11,11,0,1,11,7,11,11,9,11,10,7,2

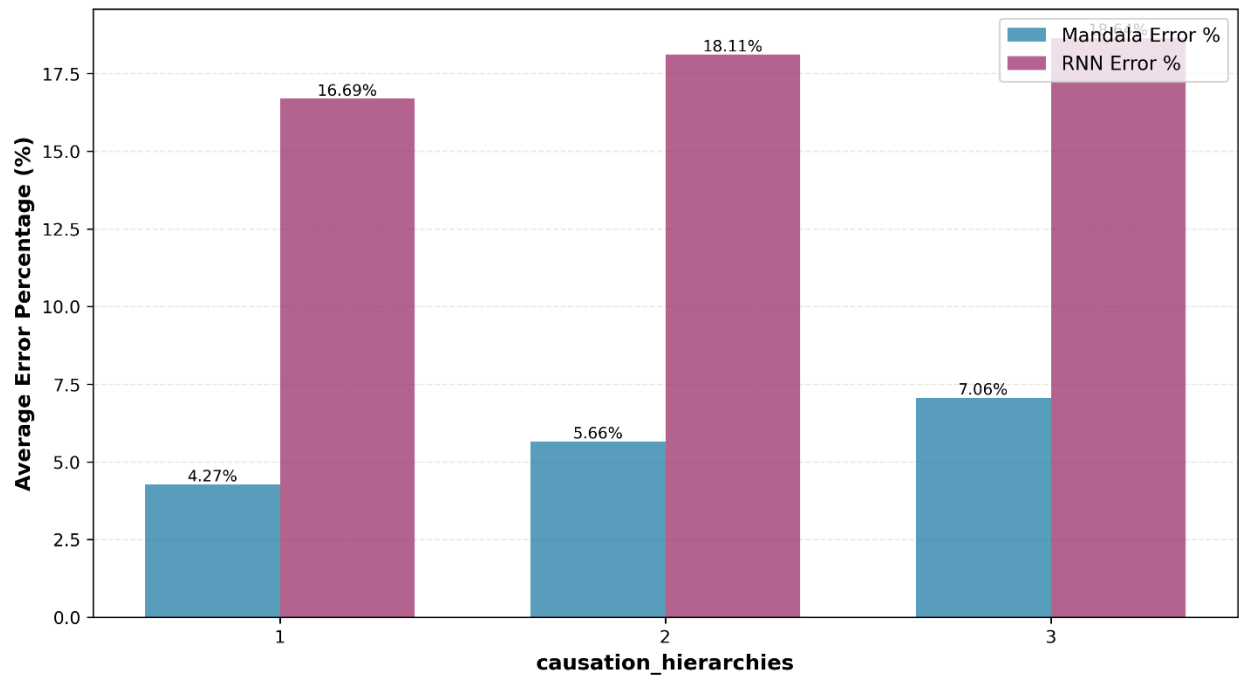
10,8,9,7,4,5,9,8,8,10,11,11,7,9,9,11,2

Note that 4,5 causes 2,3 and 0,1 causes 2,6

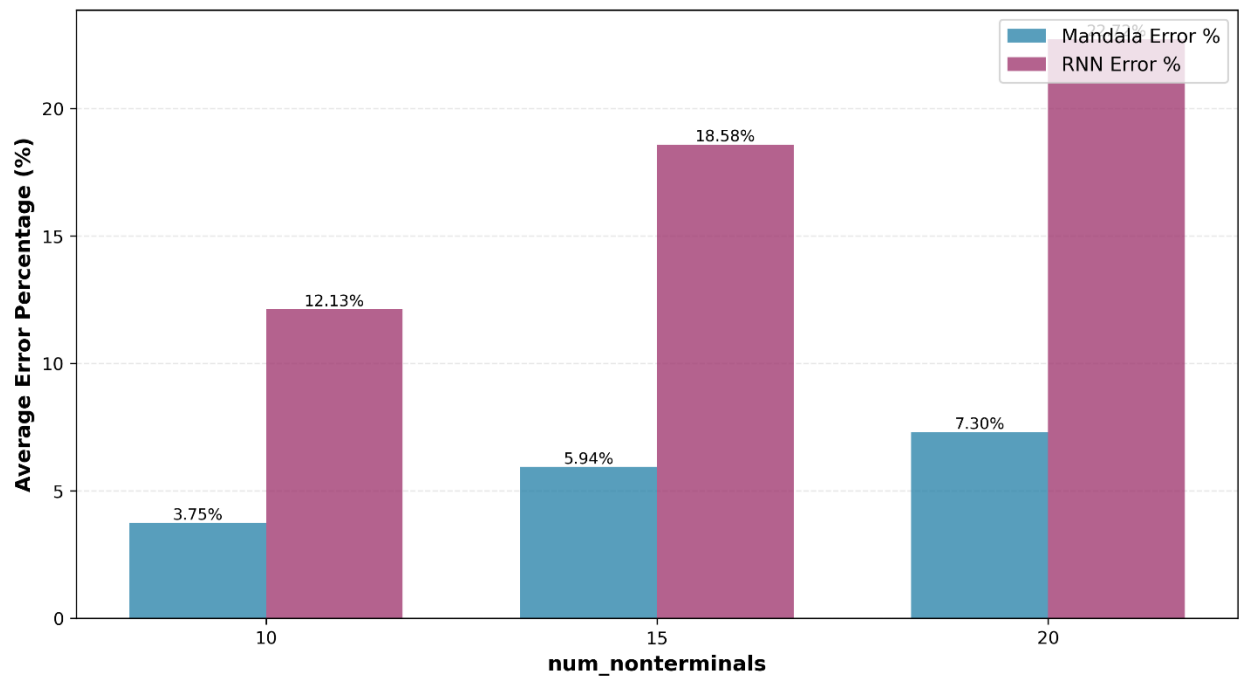
Claude predicted the first two simple sequences, But failed on the sequence that depended on recognizing the earlier transitions determine the ending transitions.

ChatGPT did not correctly predict any of the sequences.

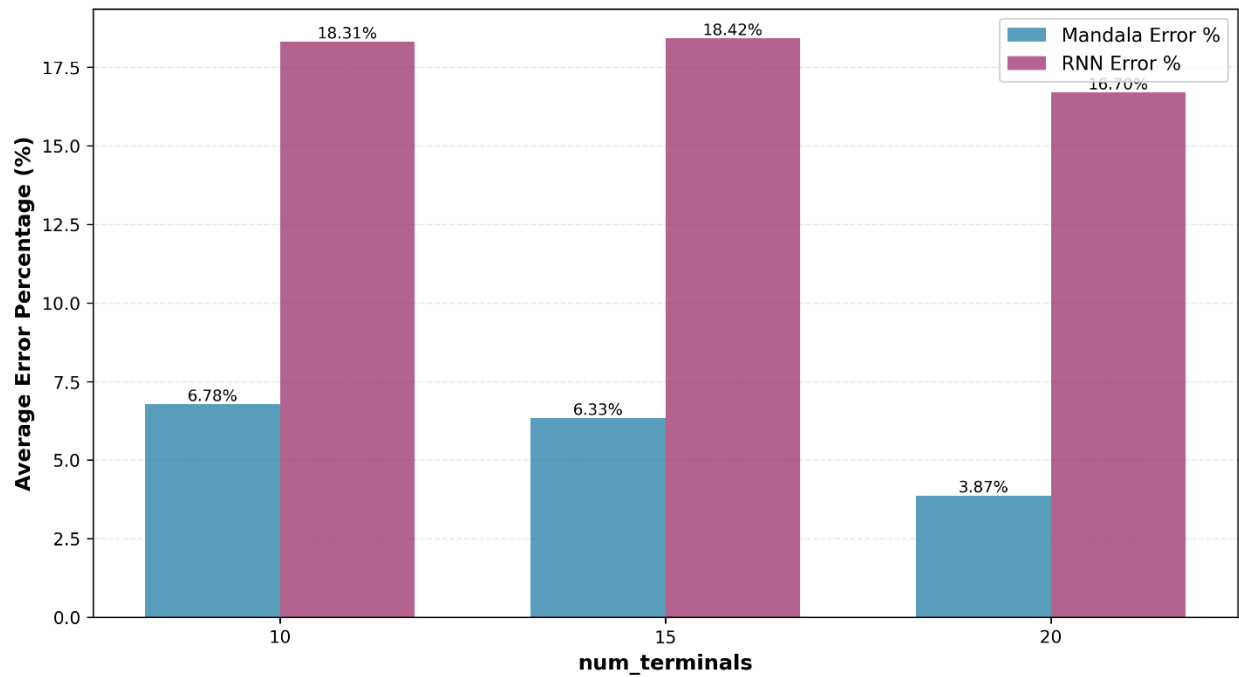
Mandala vs RNN: Average Error



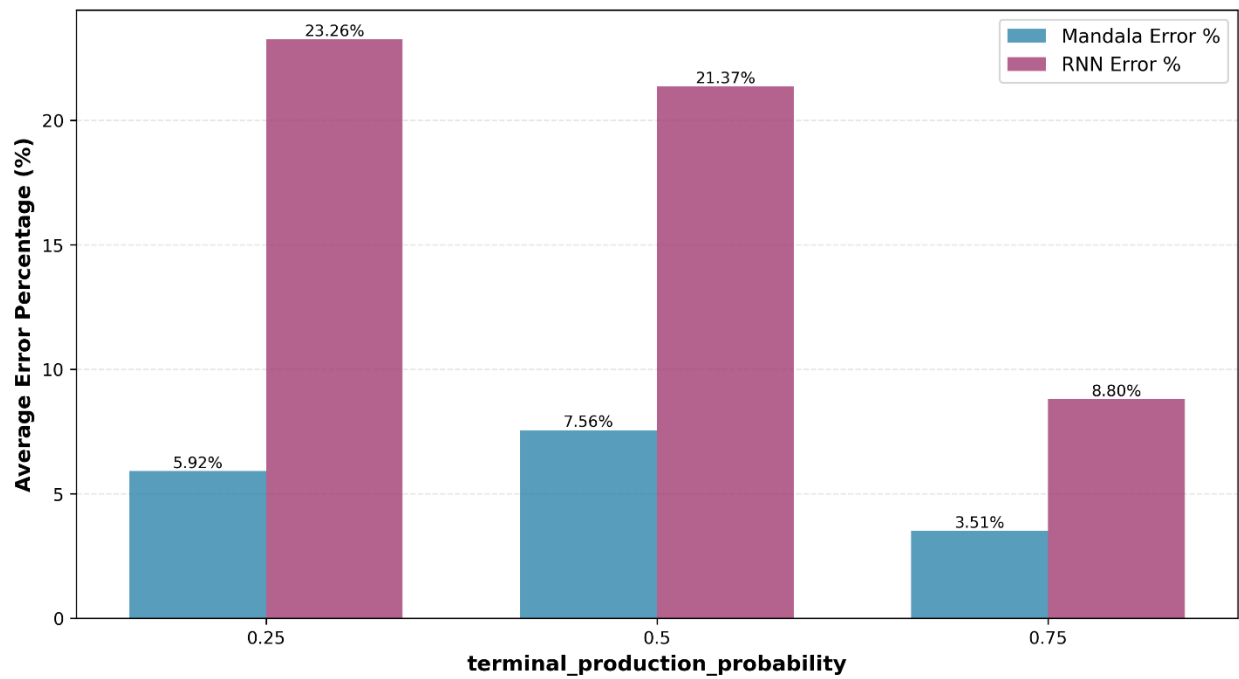
Mandala vs RNN: Average Error



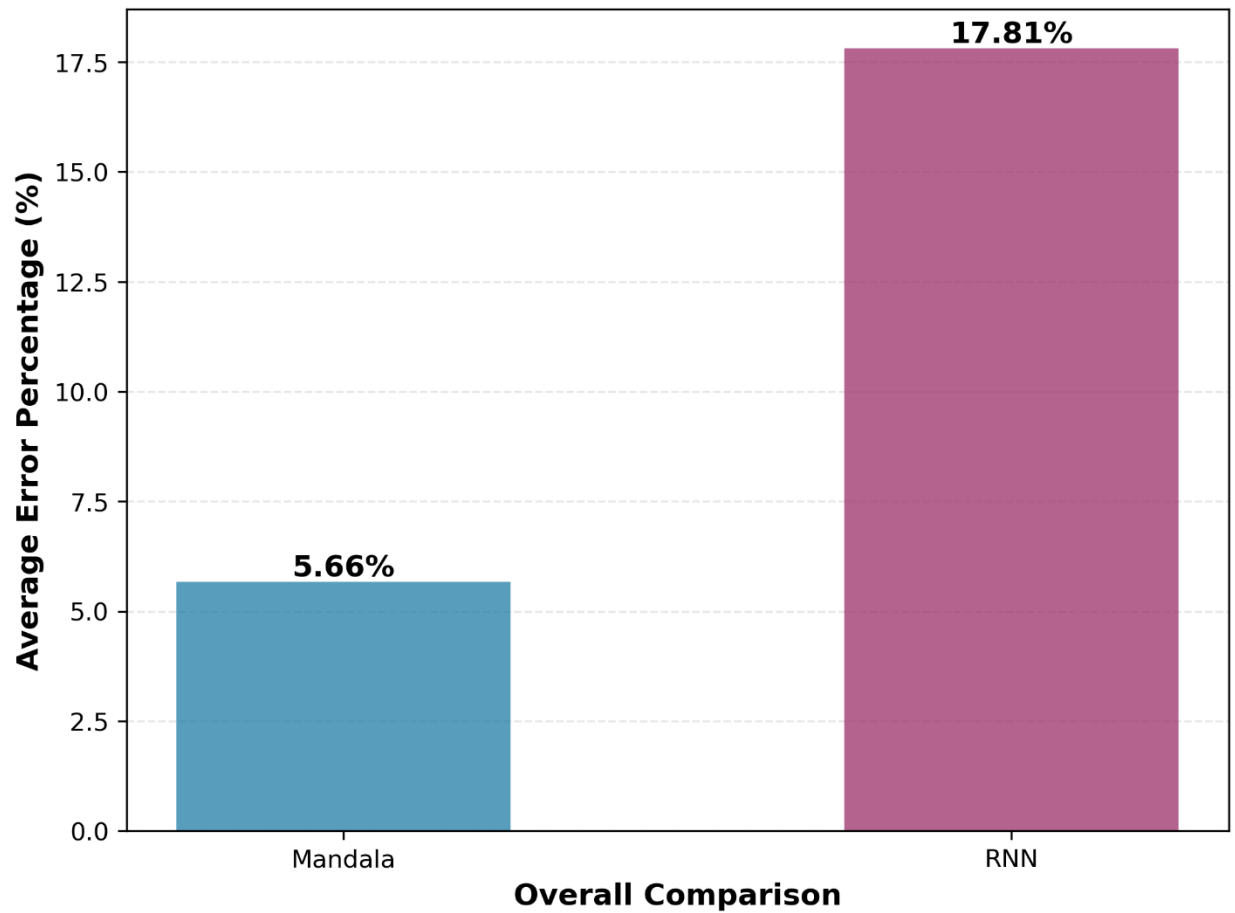
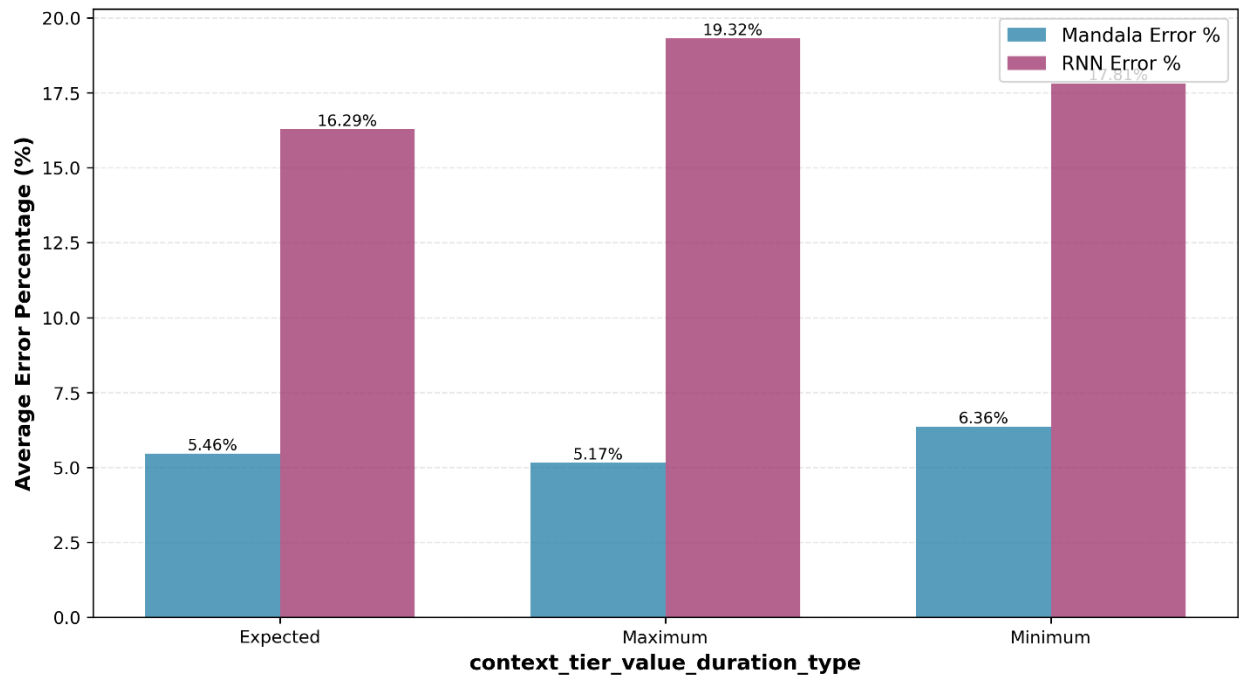
Mandala vs RNN: Average Error

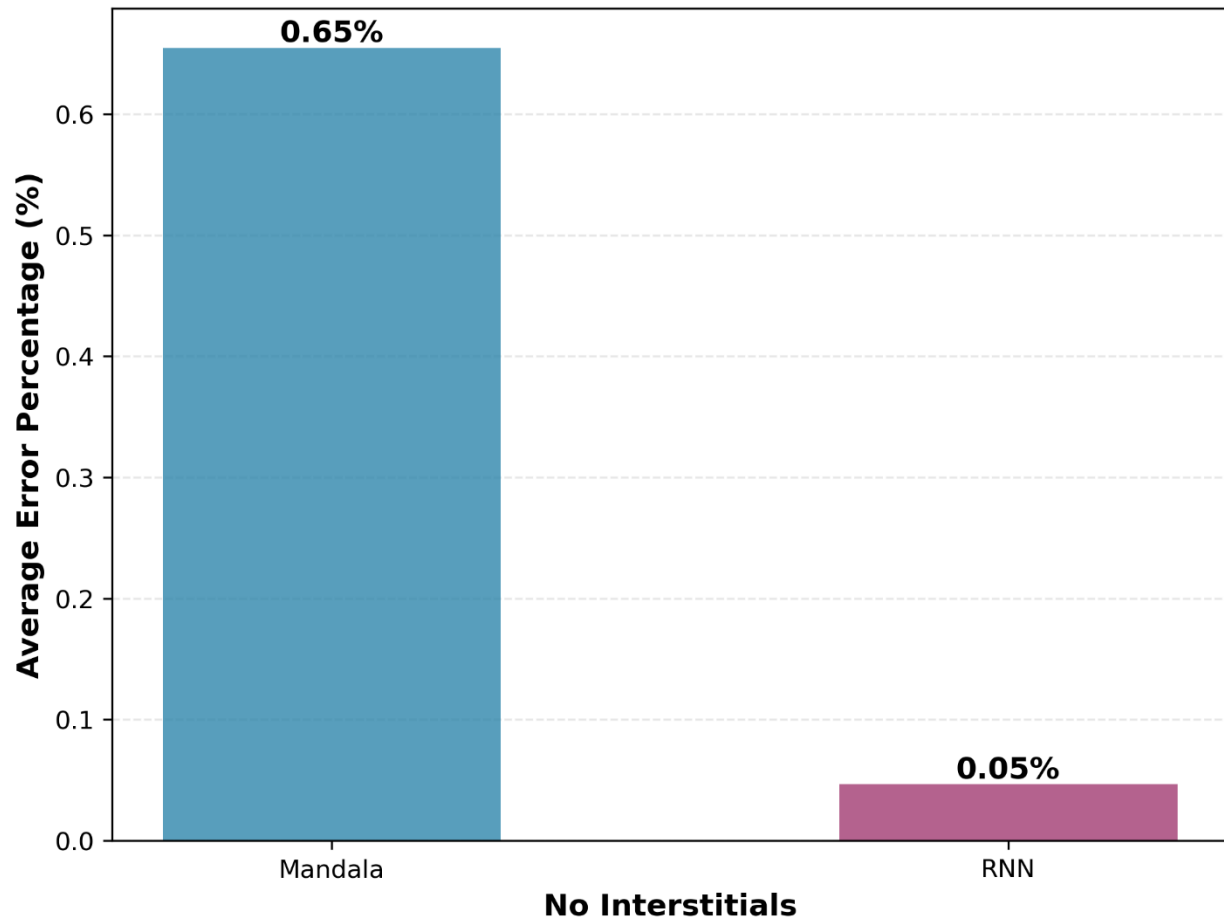


Mandala vs RNN: Average Error



Mandala vs RNN: Average Error





CONCLUSION

Note for paper: possible to learn out-of-order streams, but not done here. Reference my paper on conjunctions.

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